American Computer Science League

2022-2023 ● Contest 4: Short Problems Solutions ● Junior Division

1. Graph Theory

The adjacency matrix with double arrows everywhere is as follows:

0	1	1	1	1	1
1	0	1	0	0	1
1	1	0	1	0	0
1	0	1	0	1	0
1	0	0	1	0	1
1	1	0	0	1	0

There are 20 entries of 1. Vertex A connects to all 5 other vertices and all of the other vertices connect to 3 others so 5+ 5*3 = 20.

C. 20

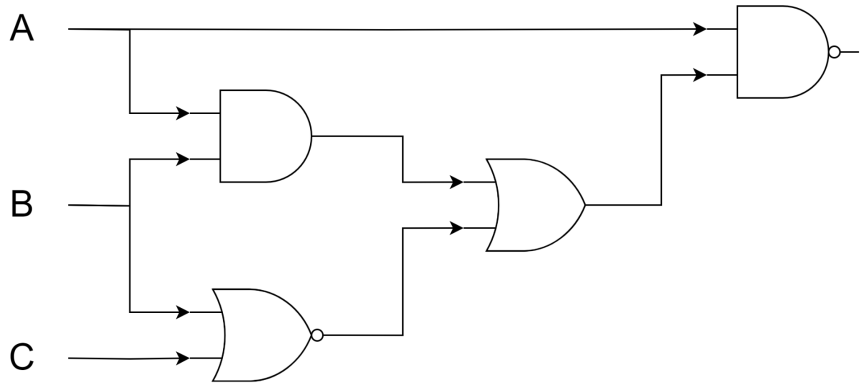
2. Graph Theory

By inspection, there are 6 of them: ABA, ABCA, ADCA, ADCBA, BCB, CDC

D. 6

3. Digital Electronics

Translate the following digital circuit into a Boolean expression.



The gates from top to bottom are NAND, AND, OR, and NOR. The lower part is (A AND B) OR NOT (B OR C). That is $AB + \overline{B + C}$. Adding the top part gives you $\overline{A(AB + \overline{B + C})}$.

A. $\overline{A\,(AB + \overline{B + C})}$

4. Digital Electronics

This digital circuit is translated to $(A + B)(\overline{(\overline{B + C})C}$. You could use a Truth Table, but simplifying the expression yields the following:

$(A + B)(\overline{(\overline{B + C})C})$
$= (A + B)(\overline{\overline{B + C} + \overline{C}})$
$= (A + B)(B + C + \overline{C})$
$= (A + B)(1)$
$= A + B$

This expression is TRUE for 6 of the 8 possible ordered triples, all but (0,0,*).

E. 6

5. What Does This Program Do? (Strings)

`s = "JavaPythonC++"`
`ans1 = s[3:]`
`ans2 = s[4:9]`
`ans3 = s[:4]`
`output ans2 + ans1 + ans3`
	This simply rearranges the 3 words and gives you the letters from position 4 to position 9, the last 3 letters of the string, and the first 4 letters of the string. Therefore it gets "PythonC++Java".

E. PythonC++Java

6. What Does This Program Do? (Strings)

```
s = "summeriscoming"
ans = ""
for x = 0 to len(s)/2 - 1
    y = len(s) - x - 1
    if s[x] < s[y] then
        ans = ans + s[x]
    else
        ans = ans + s[y]
    end if
next x
output ans
```

This program creates a string with each letter that is less than its corresponding one at the beginning and end of the string. Therefore, it's "gnimeci".

A. gnimeci